SafeSpare AI

Invest only what life can safely spare.

FinTech | Problem Statement 2: Smart Expense & Micro-Investment Assistant

Team Kryptonite

Members

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Spend | **Protect** | **Recover** | **Grow**

A safety-first finance assistant that protects essentials before savings.

Problem Statement

FinTech Problem Statement 2 - Smart Expense & Micro-Investment Assistant

Build an application that tracks spending, categorizes transactions, rounds up purchases, and simulates spare change into a portfolio with actionable insights.

The gap

Most round-up tools assume every spare rupee is safe to save. But the user may still have rent, EMI, insurance, bills, groceries, and salary timing risk ahead.

What a winning solution must show

1 **Understand spending**

Clean transactions and categorize where money goes.

2 **Protect essentials**

Reserve bills, EMI, insurance, groceries and buffers first.

3 **Grow safely**

Only safe round-ups and user-confirmed savings enter simulation.

Solution: SafeSpare AI

A safety-first micro-investment assistant

SafeSpare analyzes a user statement, identifies safe savings opportunities, and simulates progress toward goals without moving money.

Upload PDF/CSV statement → **Understand** Categorize spending → **Protect** Bills + safety buffer → **Recover** Subscriptions + price hikes → **Round up** Eligible small purchases → **Simulate** Goal + mutual fund view

Key modules

Safe Spare Engine: calculates what is truly spare after upcoming essentials and uncertainty reserve.

Leak Radar: detects recurring payments, duplicates and silent price increases, but never assumes unused without user confirmation.

Goal Simulator: combines allowed round-ups and confirmed recovery into a simulated mutual-fund style projection.

Guardrail
No real investment. No automatic cancellation. No guaranteed returns.

Tech Stack

Hybrid AI + deterministic finance engine

Frontend

React 18 + TypeScript
Vite 5.4
react-router-dom
Recharts
Hand-written CSS + design tokens

Backend

Python 3.11 + FastAPI
Pydantic validation
pdfplumber + PyMuPDF
Deterministic Decimal engine
Uvicorn

AI / Voice

Groq - Llama 3.3 70B
OpenAI - GPT-4o-mini
Gemini - 2.0 Flash
ElevenLabs TTS
Web Speech API

AWS

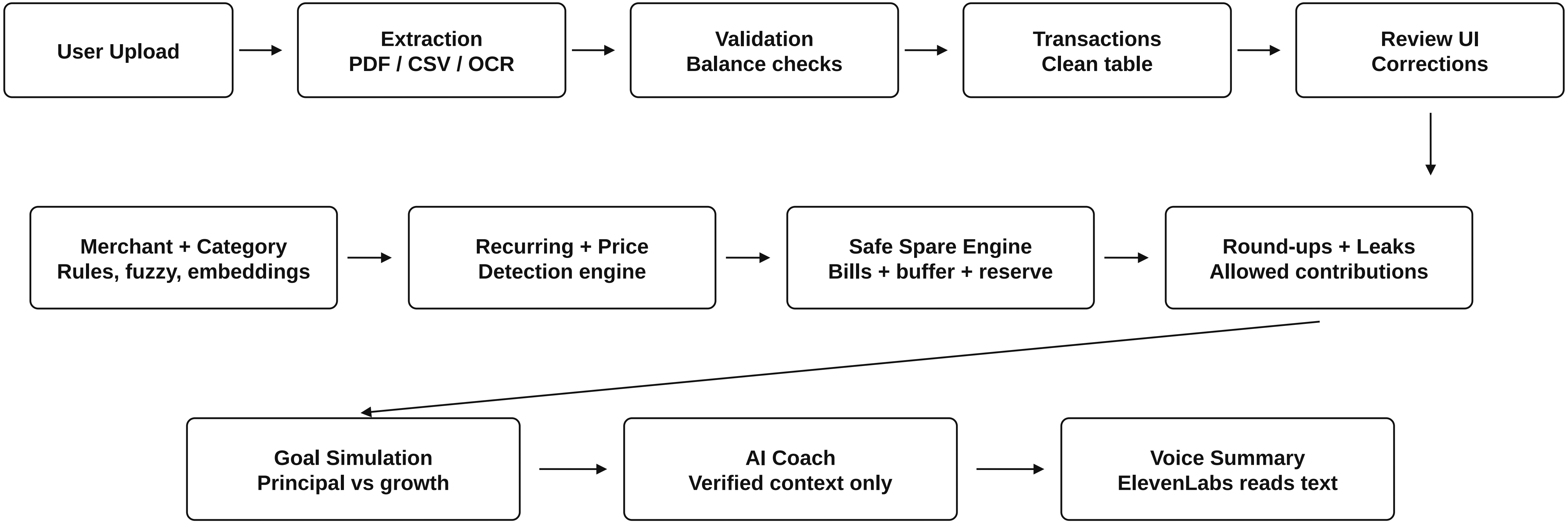
CloudFront + OAC
Private S3 static frontend
EC2 + Docker Compose
Caddy TLS + SSM Parameter Store
In-memory repositories (no database)

Accuracy principle

LLMs explain verified facts. Python calculates money. This reduces hallucination risk and keeps the output auditable.

Architecture Diagram

Evidence-first pipeline with AI guardrails



Guardrail layer: deterministic calculations, evidence trails, user confirmation, and schema-validated LLM outputs.

Links and Conclusion

Submission assets and final impact

Submission links

Deployed application	https://da2u5q8s30wam.cloudfront.net/
GitHub repository	https://github.com/Hardik182005/InnovaHack-Team-Kryptonite
Demo video	https://youtu.be/wDqa_rXf8d4

Conclusion

SafeSpare does not ask users to invest more. It first determines what their life can safely spare. It protects essentials, recovers avoidable recurring costs, and simulates responsible goal progress.

Your bills first. Your goals next.

Thank You

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